



Deconstruct artificial intelligence's productivity impact: A new technological insight

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ABSTRACT

Some viewpoints suggest that IT investment seems to fail to significantly stimulate enterprise productivity in some cases. Therefore, revealing the impact of AI on firm productivity is an important topic to analyze whether Solow's paradox can be valid in the digital age. Based on panel data of 3235 listed companies in China from 2007 to 2021, we comprehensively discuss the impact and mechanism of AI on firm productivity using fixed-effects model, systematic GMM model, and mediated-effects model. Key findings include: AI significantly improves firm productivity, especially in state-controlled, internationally minded, and innovative firms. Mitigating information asymmetry is a key channel, while specialized division of labor and independent green innovation are potential ones. Supply chain digital transformation policies enhance the productivity effect of AI, and AI shows green development benefits. Additionally, the dynamic decomposition effect shows that the productivity-enhancing effect of AI is slowing down in the long run. This research provides important insights into understanding AI's role in the digital age and holds significance for firms and policymakers.

1. Introduction

In the late 1980s, American scholar Strassman surveyed 292 enterprises and found a strange phenomenon. There was no obvious correlation between the IT investment of these enterprises and the return on investment (ROI). Robert Solow, an economist who won the Nobel Prize in 1987, called this phenomenon the "productivity paradox": "Computers everywhere except in the productivity statistics." The central argument is that despite the enormous advances made by firms in the field of science and technology, technological progress has not been translated into corresponding productivity gains throughout the economic system [1]. According to Solow, there are three broad reasons for this phenomenon. One is the delayed effect, which means that technological innovations usually take some time to fully affect productivity. The second is the problem of resource allocation. If there are problems in resource allocation by firms or Governments, this may affect the application and diffusion of new technologies, thus limiting productivity gains [2]. Thirdly, there is the issue of measurement. In emerging industries and high-technology areas, traditional productivity measurement methods may be difficult and inaccurate, resulting in a failure to scientifically reflect the benefits of technological progress. The

introduction of Solow's paradox has led to extensive discussion and research in economics, prompting greater attention to the actual impact of technological innovation on productivity and economic development. For example, green technology innovation and renewable energy technology innovation [3,4].

In fact, Solow was based on a sample of U.S. firms from 1980 to 1990, and the study found that the information and communication technology industry (ICT) seemed unable to promote productivity growth. However, in the 21st century, scholars have demonstrated that Solow's paradox can be overcome. For example, Madan and Ashok [5] state that the use of AI in government represents a fundamental shift in governance, contributing to lean government and the delivery of efficient, personalized services, and is at the forefront of governance in the digital age (DEG), even though it can be risky and ethically challenging. Building on this foundation, Madan and Ashok [6] further explored issues related to the adoption and proliferation of AI in public administration through a systematic literature review. The study points out that the implementation of AI involves technological, organizational, and environmental factors and that there are tensions between automation and augmentation, facilitation and autonomy, and data accessibility and security and privacy. Institute [7] assessed the scale and automatable

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potential of service transactions by analyzing public services and their transactions provided by the UK central government. The study found that the UK central government provides around 377 citizen services per year, that these services carry out around 965 million transactions per year, 143 million of which are complex bureaucratic processes, and that many of the tasks of the decision-oriented services have a high potential for automation. While a large number of transactions are expected to be automated in UK central government, the practical adoption of AI is lagging behind, with applications covering four themes: regulation, compliance, public service delivery and organizational management. Industry reports indicate that there is a gap between AI experimentation and benefits realization, and that successful adoption of AI requires a strategy of experimentation, innovative sourcing, and collaboration, as well as corporate sponsorship and a centralized governance team in the organization [8].

In addition, the positive role of AI in business and management is confirmed. Cost reduction and efficiency improvements are important considerations for organizations adopting AI in business and management [9]. Wu and Qin [10] proves the advantages of AI in optimizing decision-making systems for corporate control and management, and in business intelligence, AI technologies are used to collect, combine, and analyze company data for efficient allocation of business resources. Of course, AI itself does not necessarily create value for the company, the key lies in how it is used. The use of AI can be divided into two kinds: strategic and operational. operational use is to optimize the existing process in the process, but it is easy to be copied by peers, and cannot create strategic benefits [11]. Strategic use is where managers need to think about choosing the right place in the business model architecture for AI to leverage the network effect of data to attract customers, improve service, prevent customer migration and achieve sustained performance excellence [12]. Sjödin et al. [13] further explore the potential of AI for industrial manufacturers to achieve circular business model innovation and the capabilities required. AI's value discovery, realization and optimization capabilities enable manufacturers to realize economic and sustainable value.

Therefore, the first key research question and innovation of this study is to explore the impact of AI development on productivity of Chinese firms at the empirical level. The scope of this paper is the panel data formed by A-share listed companies in China from 2007 to 2021, and the focus is to explore the extent to which AI can affect firm productivity from a firm perspective. Why do we say this is the first point of innovation? This is because past research has focused on the impact of information and communication technology (ICT) on productivity at the macro-provincial or municipal level. For example, Wang and Guo [14] explored the positive effects of ICT on green total factor productivity in local cities and the negative productivity impacts on neighboring regions, using 282 cities in China as the scope of the study. Erumban [15] demonstrated the proved impact of ICT on labor productivity, with lower transaction costs as the path, using a study sample of more than 100 countries around the world from 1997 to 2017. Compared to AI, ICT is an earlier and macroscopic concept that covers both information technology and communication technology [16,17]. Different from previous studies, this study constructs a thesaurus based on crawler technology by crawling the words related to "artificial intelligence" in the annual reports of listed companies, and then summarizes the word frequencies and takes the logarithm to get the AI level of the companies. Further, we calculated the firm's productivity level using four methods such as Ordinary Least Squares (OLS), Fixed Effects (FE), Luhnberg Productivity Indicator (LP), and Optimal Productivity Indicator (OP). On this basis, this study for the first time empirically focuses the research perspective from macro-cities and countries to enterprises, proving the positive effect of AI on enterprise productivity.

Second, what factors actually play a mediating effect between AI and firm productivity? How do China's current digitalization and environmental policies affect the relationship between the two? The solution to these two key questions is the second innovation of this study. This is

because previous studies have focused on the role of factors such as human capital, operating costs, and digital transformation as channels between the two [18,19]. The mediating role of the above categories of factors is not an intuitive effect of AI, and its transmission role is doubtful. Instead, we focus on exploring the prominent role of AI in mitigating firms' information asymmetry based on AI's information properties. At the same time, we consider that AI may be able to promote social specialization and technological innovation, which are two channels we are expanding. Importantly, the key to boosting firm productivity is the efficient operation and integration of supply chains [20]. Therefore, we analyze the moderating role of supply chain pilot policies on the productivity effects of AI. Of course, if firms are affected by environmental regulations, their productivity enhancement strategies are bound to suffer [21]. Therefore, low-carbon pilot policies are also taken into account by us. In addition, we discuss in detail the heterogeneous effects of AI on firm productivity moderated by the nature of the firm, the international perspective of the leader, and the sense of innovation, enriching the heterogeneity analysis perspectives of existing studies.

In addition, there may be a temporal transmission effect of historical information infrastructure developments on current AI technology advances. Unfortunately, this potential moderating effect has not been explored. Therefore, this study aims to address this issue and presents it as the third innovation. Specifically, we innovatively consider the differential impact of historical information infrastructure development on the productivity effect of AI. We use the Ming Dynasty stagecoach as a proxy variable for the historical impact, which is justified by the fact that the stagecoach, as a major transportation route and information transit station, facilitated the transfer of information and the circulation of knowledge, and inspired people to think about intelligent management [22]. In addition, differences in the level of productivity of the firms themselves can also interfere with the impact of AI. Therefore, we further discuss the differential impact of AI on firms' productivity at different productivity levels using a panel quantile regression model (PQRM) [23]. Importantly, the impact of AI on productivity cannot be static. When discussing the impact of ICT or digital technology, previous studies tend to focus only on the static impact of both [24,25]. Instead, we further analyze the dynamic impact of AI on productivity based on a panel vector autoregressive (PVAR) model.

This study is organized as follows. Section 2 is a review of relevant literature. Section 3 is the theoretical analysis and hypothesis formulation. Section 4 is the research design, including model design, data sources and variable settings. Section 5 presents the empirical results and discussion, including benchmark regression results, robustness tests, heterogeneity analysis and mechanism tests. Section 6 provides further analysis and discussion, including policy moderation effects, green development analysis, historical impact discussion, quantile regression analysis and dynamic decomposition analysis. Section 7 is a discussion chapter that includes academic and practitioner implications, limitations and future research agenda. The final section summarizes the full text and provides targeted policy recommendations. As shown in Fig. 1, the technical route of this study is as follows.

2. Literature review

Next, we outline three categories of literature that are closely related to this study. The first category of literature is AI-related definitions and their implications. The second category of literature is the factors affecting firm productivity. The third category of literature is the study of the correlation between AI and productivity.

AI, as the name suggests, refers to technologies and methods that simulate, extend, or expand human intelligence so that computer systems have intelligent behaviors like humans. These intelligent behaviors include learning, reasoning, understanding language, recognizing objects, solving problems, and adapting to the environment. Ashok et al. [26] explored the key digital ethical implications of AI use in digital

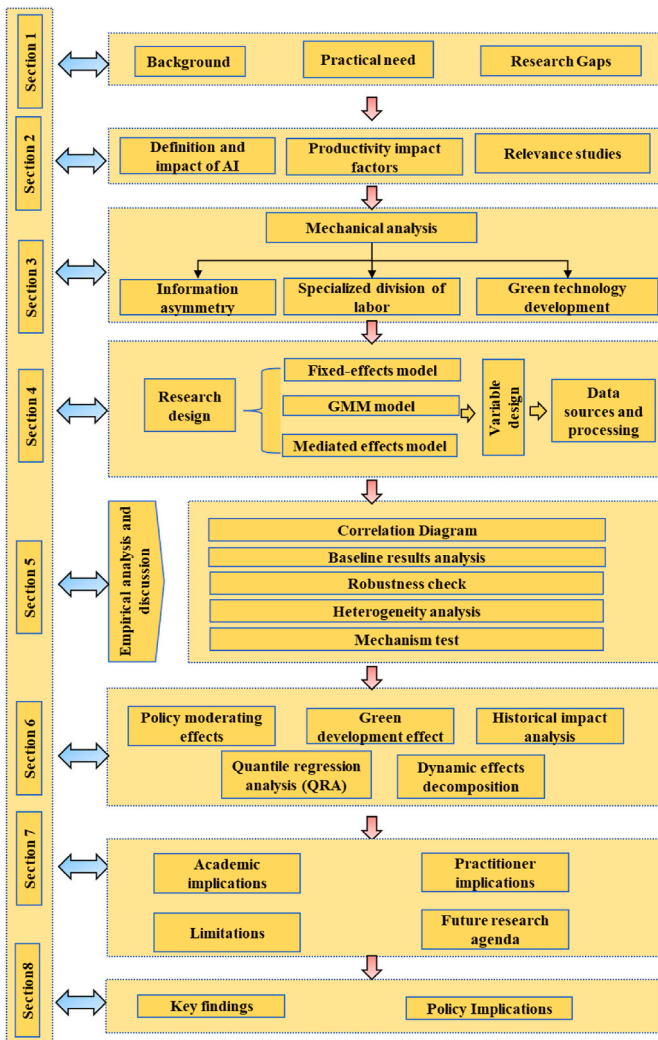


Fig. 1. Technology roadmap.

technology (DT) through a systematic literature review and qualitative synthesis. The results indicate that the ethical implications of intelligence, accountability, fairness, autonomy, and privacy were most widely discussed in the sample, while the ethical implications of the governance domain were generally applicable to most digital technology prototypes. Based on its rich literature review, this study first clarifies the meaning of "technology, digital technology and AI". Technology: Technology is broadly defined as "all tools, machines, appliances, weapons, instruments, housing, clothing, communication and transportation equipment, as well as the skills we use to produce and use them. Arthur [27] further expanded the concept of technology by arguing that technology is not limited to tangible tools. It also includes collections, practices, and processes that satisfy human needs and the aggregation of available tools. Digital Technology (DT): Berger et al. [28] define DT as "multiple principles for the use of digital resources for effectively finding, analyzing, creating, communicating, or using information in a given context". DT can be defined as "multiple principles for the use of digital resources that are used to effectively find, analyze, create, communicate, or use information in a given context", which is a subset of technology. Artificial Intelligence (AI): AI is defined in the article as a machine or "collection of technological components" that is capable of performing cognitive functions associated with the human mind, operates autonomously and without human intervention, and is capable of learning and recognizing patterns to make decisions. Specifically, AI can be conceptualized at the conceptual level as

encompassing multiple principles for automating and augmenting cognitive tasks performed by humans.

The impact of AI on modern society is broad and profound, and many scholars have analyzed the comprehensive impact of AI on the economy and society based on different industries. For example, Han and Wu [29] found that AI can help solve the dilemmas faced by the biomedical industry, such as outdated processes and bulky instruments, and in particular can advance basic translational research in the biomedical field through algorithmic modeling. Singh et al. [30] further analyzed the micro-help of AI on drug screening and concluded that AI has changed the drug development process, especially can high-performance algorithms improve the accuracy of drug design. Therefore, the use of AI technology to enhance customized patient experience, improve clinician capabilities, and improve healthcare service processes is an important healthcare industry reform direction [31]. In the banking industry, AI is also gradually leveraging its powerful tool performance, especially as AI can bring about effective innovation in business models through accurate risk assessment, quantitative trading and optimized investment decisions [32]. Additionally, the application of AI in the building industry has brought about a full lifecycle transformation. ai can improve energy efficiency through automation and intelligent management in all phases of building design, control, maintenance, and retrofitting [33].

Firm productivity is affected by a variety of factors. Enterprise productivity refers to the efficiency and ability of an enterprise to produce goods or provide services over a given period of time [34]. It is usually measured in terms of the ratio of outputs to inputs, such as the number of products produced per unit of time and the value created per unit of resources. An increase in productivity of an enterprise means that the enterprise can utilize its resources more efficiently and obtain more output with less inputs, thus improving its competitiveness and economic efficiency. Specific measurement indicators can vary according to different industries and production characteristics [35]. Policy factors are one of the factors that most directly affect firm productivity and are most frequently discussed in academia. Zhang et al. [36] use fiscal subsidies and tax incentives as two proxy variables for policy analysis to explore the productivity effects of the two types of policies on Chinese industrial firms. The results show that both types of policies have positive productivity effects, with tax incentives clearly having a stronger stimulus effect. The impact of cleaner production on firms' production performance is also evident. Cleaner production practices by firms enhance productivity through increased demand for highly skilled labor, technological advances and equipment upgrades [37]. The role of environmental regulation is also not negligible. Wang et al. [4] used China's environmental tax as a representative of environmental regulation, and used the DID model to assess the role of environmental tax reform on firm productivity. The results found that environmental taxes hinder substantial innovation of firms and are not conducive to productivity enhancement.

With the advent of the digital era, scholars have begun to focus on the impact of digital development-related technologies on the sustainable development of cities or firms. For example, Sun et al. [3], based on panel data from 201 cities in China, demonstrate the positive impact of digital finance on carbon productivity, with human capital and marketization as important channels. In addition, digital finance, which relies on digital technology, also has an uplifting effect on carbon productivity in neighboring places. Zhang et al. [36] used the Yangtze River Economic Belt in China as a research sample to demonstrate the positive effects of synergizing green finance and high technological innovation for sustainable development. Suo et al. [38] argued that the relationship between the digital economy and firm productivity is not purely linear, and that the productivity effect of the digital economy shows an inverted U-shape, with the industry environment having a moderating effect. There is limited research on the correlation between AI and productivity. Zhai and Liu [19] explore the productivity-enhancing effects of AI technological innovations, which are greater in SOEs and

labor-intensive firms. However, Chang et al. [39] find that the impact of AI and natural resource markets on productivity is uncertain, with both positive and negative effects. In particular, the use of industrial robots can significantly reduce carbon emissions. In addition, scholars seem to be more interested in the role of AI in firms' environmental performance. Cheng et al. [40], based on a research sample of Chinese firms, found that AI can enhance firms' environmental protection capabilities, and TFP is an important channel. Zhang and Zeng [41] investigated the impact of AI on energy intensity based on data from listed companies in China. They find that AI significantly suppresses energy intensity, especially affecting energy-intensive firms and state-owned enterprises.

This study aims to fill a number of research gaps that exist in the current literature and have an important business need. The existing literature is deficient in terms of the impact of AI on business in the following areas:

Existing literature on the impact analysis of AI mostly focuses on the macro national and city level [42,43], and lacks an in-depth exploration of the impact on the actual development of enterprises. However, enterprises are the main body of economic development, and understanding the specific impact of AI on enterprise productivity is crucial for enterprises to formulate strategies and make decisions. This study can more accurately assess the micro-enabling effect of AI on enterprises and help them better apply AI technology to improve productivity and competitiveness. In short, demonstrating the generalized productivity-enhancing effect of AI development in Chinese listed companies over the period 2007–2021 is the first business requirement.

There are numerous factors affecting firm productivity, and while traditional research has focused on policy, environmental regulation and finance [2,44], relatively little attention has been paid to emerging technological forces, such as artificial intelligence. With the rapid development of AI technology, its impact on firm productivity has become increasingly significant. Therefore, there is a need for an in-depth study of the heterogeneous impact of the emerging technology of artificial intelligence on firm productivity. In particular, this study comprehensively evaluates the variability of different types of firms, such as SOEs, non-SOEs, internationally oriented firms, non-internationally oriented firms, innovation-minded firms, and innovation-unaware firms, in the uptake and utilization of AI technologies to achieve productivity gains. The variability in the adoption of AI-enabled productivity improvement in the above six different types of organizations is then the second business need of this study.

The existing literature focuses more on the broader impacts of the digital economy and less on the specific role of AI in firm productivity [45,46]. This study will focus on the impact of AI on firm productivity, provide a comprehensive and quantitative assessment through multiple econometric models, and deeply analyze the mechanisms behind it. Specifically, through theoretical elaboration and empirical analysis, this study more comprehensively verifies the transmission roles of information asymmetry, social specialization division of labor, and green technological innovation in AI-enhanced firm productivity. Exploring the indirect driving effect of AI on firm productivity through the above three channels and the differential impact of these three channels is the third business need of this study.

3. Theoretical analysis and hypothesis formulation

First of all, we would like to discuss the reasons why AI works directly on enterprise production and improves its productivity from multiple perspectives. Overall, AI plays an important role in improving enterprise productivity and can help enterprises achieve higher levels of automation, intelligence and personalized production, thereby increasing competitiveness and gaining greater market share [47].

Specifically, AI can directly empower enterprise production in five ways. First, AI can promote the automation and intelligent upgrading of enterprise production. AI technology can be used to automate repetitive tasks and processes, thereby reducing the need for human resources and

lowering the error rate. Through intelligent production processes, enterprises can manage resources and time more efficiently [48]. Second, AI has powerful data analysis and prediction functions. AI can analyze large amounts of data and extract valuable information to help enterprises make more accurate predictions and decisions. By analyzing market trends, customer demand and supply chain data, companies can better plan production schedules and resource allocation to improve productivity [49]. Third, AI can help companies with quality control and preventive maintenance. AI technology can monitor key parameters in the production process in real time and identify potential problems or anomalies in a timely manner. Through real-time monitoring and predictive maintenance, companies can reduce equipment downtime, improve productivity, and ensure that product quality meets standards [50]. Fourth, AI can provide personalized production and customization services. AI can realize personalized production according to customer needs and preferences, thus improving the market competitiveness of products. Through the intelligent production system, enterprises can adjust their production lines more flexibly, respond quickly to market changes, and meet the personalized needs of customers [51]. Fifth, AI has human resource optimization capabilities. AI technology can help enterprises optimize the allocation and management of human resources and improve employee efficiency and satisfaction. Through automation and intelligent tools, enterprises can reduce the investment of human resources in low value-added tasks, thus allowing employees to focus more on innovation and high-value work [4]. In summary, through literature analysis and theoretical elaboration, AI can promote automation and intelligent upgrading of enterprise production, provide powerful data analysis and prediction functions, facilitate quality control and preventive maintenance, realize personalized production and customized services, and optimize human resources, thus directly enhancing enterprise productivity. In short, Hypothesis 1 can be formulated and confirmed pending empirical analysis in Section 5.

H1. AI has a direct and positive impact on firm productivity.

It is important that we consider the main channels through which AI improves productivity. The first and primary channel is to mitigate information asymmetry. Information asymmetry is a market in which there is an information discrepancy between buyers and sellers that results in unequal or inefficient transactions [52]. Given AI's unique advantage in processing massive amounts of information, it is irreplaceable in alleviating enterprise information asymmetry. First, AI can help enterprises predict market demand more accurately by analyzing big data to discover potential market trends and consumer behavior patterns. This prediction capability can reduce the uncertainty of market information and lower the degree of information asymmetry for enterprises [53]. Second, AI can personalize marketing and services based on customers' historical behaviors and preferences, improving the match between products and services. This personalization capability can increase customers' trust and loyalty to the company and reduce the impact of information asymmetry [54]. Third, AI can monitor the market and customer feedback in real time and adjust production plans and marketing strategies in a timely manner. This real-time monitoring and feedback mechanism can reduce the lag in the transmission of information and reduce the degree of information asymmetry [55]. Finally, AI can optimize supply chain management and improve the efficiency and reliability of production and distribution. By monitoring and forecasting all aspects of the supply chain in real time, AI can reduce information asymmetry caused by supply chain problems [56]. In summary, AI improves productivity by alleviating information asymmetry, specifically by analyzing big data to more accurately predict market demand, personalize marketing and services, monitor market and customer feedback in real time, and optimize supply chain management. Thus, hypothesis 2a can be formulated.

H2a. AI improves firm productivity by reducing information asymmetry.

In addition, two other channels that we need to consider are the division of social specialization and green innovation, respectively. In terms of social specialization and division of labor, on the one hand, AI can be used to automate specific tasks, thereby reducing the need for human resources. Through automation, companies can delegate repetitive and low value-added tasks to AI systems, allowing human resources to focus on higher-value work and increase productivity [57]. On the other hand, AI enables intelligent collaboration and optimization between machines to improve the overall efficiency of production lines. By connecting multiple intelligent systems for data sharing and collaborative work, companies can achieve higher levels of specialized division of labor and collaborative efficiency [58]. In summary, hypothesis 2b can be formulated.

H2b. AI can enhance business productivity by facilitating the division of social specialization.

As far as green innovation is concerned, on the one hand, AI can facilitate the development and innovation of green technologies and help enterprises develop more environmentally friendly products and production processes. By introducing new environmentally friendly technologies and solutions, companies can improve the market competitiveness of their products and enhance production efficiency [59]. On the other hand, AI can promote green innovation by increasing employee awareness and participation in environmental protection. Through intelligent tools and training, employees can better understand and practice environmental concepts and thus actively participate in the green innovation and production process [60]. Based on the above analysis, hypothesis 2c can be formulated.

Hypothesis 2c. AI can enhance firm productivity by promoting green innovation.

Based on the above analysis, the impact mechanism of this study for AI and productivity is shown in Fig. 2.

4. Methodology

4.1. Model setup

Based on the analysis of the theoretical framework, we first develop a multidimensional fixed effects model to assess the direct impact of AI on firm productivity. The model is shown in Eq. (1).

$$TFP_{ijt} = \alpha + \beta AI_{ijt} + \gamma X_{ijt} + \mu_i + \tau_j + \delta_t + \varepsilon_{ijt} \quad (1)$$

Where TFP_{ijt} represents the productivity of firm i in industry j at point t . α is the intercept term; β is the coefficient to be estimated in this study, which represents the direct effect of AI on firm productivity. AI_{ijt}

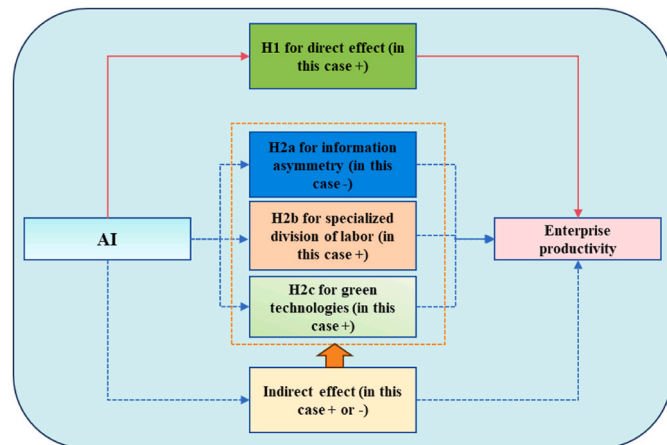


Fig. 2. Theoretical mechanisms.

represents the level of artificial intelligence of the corresponding firm. γ is the coefficient affecting the control variable; X_{ijt} represents the control variable. μ_i stands for firm fixed effects; τ_j denotes industry fixed effects; δ_t denotes year fixed effects. ε_{ijt} stands for the random perturbation term.

Further, we consider that firm productivity is most likely to be time-transmissible, i.e., productivity in the current period is easily influenced by productivity in the previous period. Therefore, we choose a dynamic panel model to further accurately estimate the effect of AI on productivity. It is worth mentioning that System Generalized Moment Estimation (SYS-GMM) not only estimates time-locked effects, but also has advantages in overcoming the endogeneity problem. Based on the above idea, the SYS-GMM model is set up as follows.

$$TFP_{ijt} = \alpha + \theta TFP_{ijt-1} + \beta AI_{ijt} + \gamma X_{ijt} + \mu_i + \tau_j + \delta_t + \varepsilon_{ijt} \quad (2)$$

Where TFP_{ijt-1} denotes the lagged one period of firm productivity to capture time-dynamic effects. θ is the parameter to be estimated. The other variables have the same meaning as in Eq. (1).

In order to test hypothesis 2 in the theoretical analysis, we need to construct a reasonable mechanism test model to test the indirect effect of AI on productivity. Based on Alesina and Zhuravskaya [61]'s mechanism testing model published in the American Economic Review in 2011, our mechanism testing model is set up as follows.

$$M_{ijt} = \alpha + \beta AI_{ijt} + \gamma X_{ijt} + \mu_i + \tau_j + \delta_t + \varepsilon_{ijt} \quad (3)$$

$$TFP_{ijt} = \alpha + \beta AI_{ijt} + \theta M_{ijt} + \gamma X_{ijt} + \mu_i + \tau_j + \delta_t + \varepsilon_{ijt} \quad (4)$$

Where M_{ijt} denotes the mechanism variable selected for this study. First, we begin by testing the direct effect of AI on the mechanism variables. Further, we include both the mechanism variable and AI in Eq. (4) and compare the magnitude of the estimated parameters to determine which mechanism variable is the one that primarily carries the channeling role of AI on productivity.

4.2. Variables' definition

4.2.1. Dependent variable

The explanatory variable of this study is firm productivity (TFP) [62]. Total Factor Productivity (TFP) is an important indicator of the overall productivity of a firm, and there are various methods for calculating TFP, including Ordinary Least Squares (OLS), Fixed Effects (FE), Luenberger Productivity Indicator (LP), Optimal Productivity Indicator (OP) and four other common methods. This study uses the FE method for estimation. The FE model is based on panel data, where the same set of firms are observed over a period of time and their production data are collected. The FE model assumes the existence of fixed, but unobserved, characteristics of each firm, which may affect the firm's productivity. Thus, the FE model models individual differences within firms through fixed effects terms, thus providing more accurate productivity estimates. Importantly, because the FE model controls for within-firm fixed effects, it is able to reduce the impact of omitted variable bias. This contributes to the accuracy and robustness of the estimates. In addition, the FE model retains the effect of time-invariant factors on the comparisons and therefore better explains the relationship between productivity changes within firms and changes in the external environment.

4.2.2. Independent variable

The explanatory variable of this study is the level of enterprise artificial intelligence development (AI). Chinese companies have not disclosed their AI development level, so we cannot describe the AI development level of companies through accurate data. Fortunately, the annual reports of listed companies will state the current status of their AI development in the form of text. Therefore, we build a thesaurus based on crawler technology by crawling the words related to "artificial intelligence" in the annual reports of listed companies (Fig. 3), and then

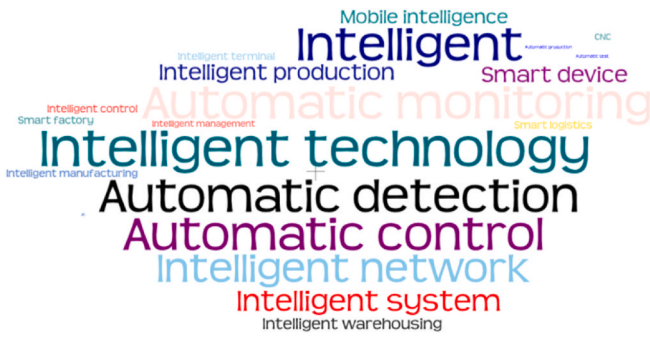


Fig. 3. AI thesaurus.

sum up the word frequencies and take the logarithm to get the AI level of enterprises.

4.2.3. Control variables

Based on previous studies [63], the following control variables were selected in this study to help exclude other factors from confounding the relationship of the variables, thus improving the accuracy and explanatory power of the model. Firm size (*AST*) is expressed as the logarithm of total assets. Return on Assets (*ROA*) is the ratio of net income to total assets. Cash reserve (*CASH*) is funds plus trading financial assets as a percentage of total assets. The gearing ratio (*TL*) is expressed as the share of liabilities as a percentage of total assets. Total Asset Growth Rate (*TAGR*) is a measure of the percentage increase in the total asset size of a business or organization over time. Shareholding concentration (*TOP*) is expressed as the percentage of shares held by the largest shareholder. Segregation of shares (*SEP*) is the percentage of shares held by the controller.

4.2.4. Intermediary variables

Based on the analysis of Hypothesis 2, we test the intrinsic mechanism of AI acting on firm productivity in this section. The mechanism variables for this study are shown below.

- (1) Information Asymmetry (*ASY*). The IA indicators in this study are based on the liquidity ratio *LR* [64], the illiquidity ratio *ILL* [65], and the yield inversion indicator *GAM* [66], and are obtained by principal component analysis. The selection of the liquidity ratio *LR*, the illiquidity ratio *ILL* and the yield inversion measure *GAM* as indicators of information asymmetry is a common practice based on relevant academic research [67]. These indicators can reflect the trading activity, liquidity situation and abnormal changes in returns in the market, thus indirectly reflecting the degree of information asymmetry. Through principal component analysis, the information from these indicators can be synthesized to provide a more comprehensive measure of information asymmetry.
- (2) Specialized division of labor (*SD*). *SD* is measured using the value-added method to indicate the extent to which an enterprise influences the division of labor in the industrial chain. The calculation method is shown in Eq. (5). The value-added method is used to measure the division of specialization because it can reflect an enterprise's ability to process and transform resources in the production process, thereby reflecting its position in the industrial chain and the extent of its influence on the division of labor. This method has been widely used in research in related fields and has a certain degree of academic rationality [68].
- (3) Green Innovation (*GI*). *GI* is represented by the number of green patents independently filed by firms in the year. The number of green patents independently filed by enterprises is used to indicate green innovation because green patents can directly reflect the innovative achievements of enterprises in environmental

protection and sustainable development [69]. Green innovation is of great significance to the sustainable development and competitiveness of enterprises, so it is reasonable to choose the number of green patents as an indicator [16].

$$SD = \frac{\text{Value added} - \text{net profit after tax} + \text{normal profit}}{\text{Income from operations} - \text{net profit after tax} + \text{normal profit}} \quad (5)$$

4.3. Data

Limited to data availability and in order to ensure the continuity of the sample, this study compiles and obtains the panel data of China's Shanghai and Shenzhen A-share listed companies from 2007 to 2021 for empirical analysis. The A-share listed companies in Shanghai and Shenzhen, China are chosen for this study because of the importance of these companies in the economic system and the relative completeness and availability of their data. During the sample period, we have tried to cover all eligible listed companies as far as possible to ensure the representativeness of the sample. The exact sample size will depend on the actual situation of the data, i.e. all companies listed during the time period are included. We ensure the quality and reliability of the sample through specific screening criteria, such as excluding companies with incomplete or unusual data. Specifically, this study further processes the raw data obtained as follows: (1) We exclude companies with serious missing key indicators. (2) Companies facing termination of listing or about to be delisted were removed from this study. (3) For outliers, we use 1 % and 99 % shrinkage. (4) For non-ratio variables, we take logarithms to try to eliminate the heteroscedasticity problem. According to the above processing, we obtained unbalanced panel data from 3235 companies in China. The data for the explanatory variables in this paper come from *CSMAR* database and *WIND* database. The data for the explanatory variables are obtained from the annual reports of listed companies. The data for control variables are obtained from *CSMAR* database and *RESSET* financial database. Table 1 reports the results of descriptive statistics.

5. Empirical findings

5.1. Correlation tests

Correlation tests can help us understand the relationship between variables. If there is a high correlation between two independent variables, it may mean that they provide similar information, which may affect the interpretation and robustness of the results of the regression analysis. Multicollinearity may lead to inaccuracy and instability in the estimation of regression coefficients. By testing for multicollinearity, the degree of association between independent variables can be assessed, and potential covariance problems can be detected early and resolved to ensure the accuracy and robustness of regression coefficients. Table 2 reports the correlations between the variables as well as their variance inflation factor (*VIF*) and *VIF/1* values. It can be seen that most of the correlations between the variables are low to moderate and no very strong correlations are observed. The *VIF* values for all variables are within acceptable limits (less than 10), indicating no serious multicollinearity problems. The relative *VIF* values are also all close to 1, indicating that the covariance between the variables is relatively weak. Taken together, the correlation and multicollinearity problems between the variables are not serious, and further regression analysis can be continued.

Further, we would like to first approximate the relationship between AI and firm productivity by using marginal histograms and scatter count plots. Both the Marginal Histogram and the Scatter Count Plot are visualization tools used to explore the relationships between variables in a dataset. Fig. 4 combines the histograms and scatter plots for AI and TFP are shown together, with the marginal histograms for the two variables shown above and to the right of the scatter plots, respectively. We

Table 1
Descriptive statistics.

Definition	Variable	N	Mean	p50	SD	Min	Max
Firm productivity	TFP	15213	2.509	2.502	0.100	2.293	2.768
Artificial intelligence	AI	15213	1.669	1.609	1.242	0.000	4.875
Firm size	AST	15213	22.120	21.950	1.213	19.980	25.970
Return on Assets	ROA	15213	0.038	0.041	0.071	−0.340	0.197
Cash reserve	CASH	15213	0.202	0.166	0.136	0.022	0.659
Gearing ratio	TL	15213	0.398	0.390	0.185	0.063	0.859
Total asset Growth rate	TAGR	15213	0.144	0.091	0.242	−0.315	1.344
Shareholding concentration	TOP	15213	0.335	0.311	0.143	0.092	0.726
Segregation of shares	SEP	15213	0.046	0.000	0.073	0.000	0.277

Table 2
Correlation matrix.

Variables	(1) AI	(2) AST	(3) ROA	(4) CASH	(5) TL	(6) TAGR	(7) TOP	(8) SEP	VIF	VIF/1
(1) AI	1								1.02	0.981276
(2) AST	0.025 ^a	1							1.4	0.714412
(3) ROA	−0.026 ^a	0.048 ^a	1						1.4	0.713744
(4) CASH	0.054 ^a	−0.201 ^a	0.244 ^a	1					1.24	0.805485
(5) TL	−0.007	0.470 ^a	−0.328 ^a	−0.416 ^a	1				1.71	0.585356
(6) TAGR	0.042 ^a	0.048 ^a	0.329 ^a	0.071 ^a	−0.006	1			1.15	0.869825
(7) TOP	−0.091 ^a	0.164 ^a	0.149 ^a	0.049 ^a	0.028 ^a	−0.018	1		1.09	0.915589
(8) SEP	−0.039 ^a	0.091 ^a	0.045 ^a	−0.009	0.059 ^a	−0.015	0.171 ^a	1	1.04	0.963866

^a shows significance at the 0.01 level.

Marginal Histogram - Scatter Count plot
mpg: lntfp vs. lnAI

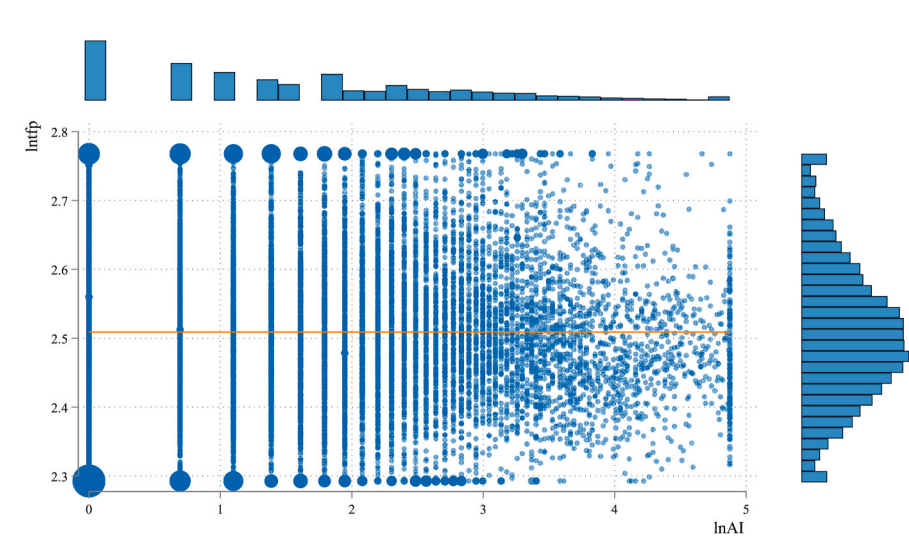


Fig. 4. Distribution of ai and TFP

counted the density distribution of data points using Scatter Count Plot. By looking at the color or size coding, we can find out the degree of aggregation and distribution pattern of data points in the scatter plot. It is not difficult to see that the TFP level of Chinese enterprises conforms to the law of normal distribution and is relatively evenly distributed. However, there are large differences in the level of AI development of individual listed companies, and their distribution is mostly concentrated in the lower-middle level. This also leads to the fact that the relationship between AI and TFP is not clear and needs to be further analyzed through precise econometric modeling.

5.2. Benchmark results

Based on Hypothesis 1, we first test the direct effect of AI on productivity using model (1), and the results are shown in Table 3. Column

(1) is the result without adding control variables, column (2) is the result without controlling for time effects, and column (3) is the result with both control variables added and controlling for time, industry, and firm. As expected from Hypothesis 1, the coefficients of AI in the results all passed the significance level test and showed a positive impact. This suggests that the use of AI in business can contribute to productivity improvement. Consistent with the findings of Wang et al. [16], they demonstrate that energy firms' adoption of AI can enhance their innovation efficiency. Carrying on their study, we extend our study to all Chinese listed companies that are accessible, further demonstrating the productivity dividend effect of AI. The reason for this is that AI's strengths in automation and intelligence can help companies make smart decisions and optimize production processes. By analyzing big data, predicting trends and simulating different scenarios, AI can provide companies with more accurate and faster decision support. For

Table 3
Baseline regression.

	(1)	(2)	(3)
<i>AI</i>	0.0047*** (0.000)	0.0015*** (0.000)	0.0007** (0.000)
<i>AST</i>		0.0644*** (0.001)	0.0639*** (0.001)
<i>ROA</i>		0.1429*** (0.007)	0.1403*** (0.007)
<i>CASH</i>		−0.0172*** (0.003)	−0.0241*** (0.003)
<i>TL</i>		0.0501*** (0.003)	0.0441*** (0.003)
<i>TAGR</i>		−0.0191*** (0.001)	−0.0163*** (0.001)
<i>TOP</i>		−0.0218*** (0.005)	−0.0162*** (0.005)
<i>SEP</i>		−0.0038 (0.006)	−0.0025 (0.006)
_cons	2.5021*** (0.001)	1.0695*** (0.015)	1.0845*** (0.019)
Year FE	✓	×	✓
Industry FE	✓	✓	✓
Firm FE	✓	✓	✓
Obs.	14616	14616	14616
R ²	0.9254	0.9693	0.9703

Notes: Numbers in () are robust-heteroskedasticity standard errors; *, **, and *** signify the significance levels of 0.10, 0.05, and 0.01, respectively. Following table is the same.

example, in production plan optimization, inventory management and supply chain adjustment, AI can help companies better respond to changes in market demand and improve productivity and flexibility [70, 71].

Further, we observe the regression results for the control variables. It can be seen that *AST* can significantly increase productivity. Larger firms are usually able to enjoy more economic scale advantages, including division of labor, resource sharing, and purchasing advantages. These advantages can reduce the production cost per unit of product and increase productivity. *ROA* also has a significant positive effect on productivity. This suggests that a high *ROA* may reflect the fact that firms utilize their assets more efficiently and avoid idle or wasted assets. Fully utilizing asset resources can increase productivity and output levels, thereby increasing productivity. *TL* also has a significant productivity-enhancing effect. A moderate gearing ratio can help firms to flexibly raise capital for investment in production equipment renewal, technology upgrading and market expansion. This helps to increase productivity and competitiveness. Unfortunately, *CASH*, *TAGR* and *TOP* have a dampening effect on productivity. This suggests that excessive cash reserves may lead to inefficient utilization of funds by firms. Funds are left idle in cash reserves instead of being used to invest in production facilities, R&D innovations, or market expansion, which may lead to a decline in productivity. In addition, too rapid a growth in total assets may lead to a fragmentation of corporate resources and an inability to focus on core business and key projects, thus affecting productivity and the degree of specialization of the enterprise. Finally, excessive equity concentration may increase corporate governance risks, which may, for example, lead to potential abuse of power, internal corruption and shareholder rights protection issues, thus affecting productivity and long-term stability of the firm. *SEP* has not yet had a significant impact on productivity. In summary, hypothesis 1 is confirmed.

5.3. Robustness testing

We use two ways to robust the findings, the first is to change the dimensionality of the fixed effects. Column (1) in Table 4 shows the estimation results after adding city fixed effects. Controlling for city fixed effects removes interference from specific influences that may exist across cities. For example, factors such as the level of economic

Table 4
Robustness test.

	(1)	(2)	(3)
<i>AI</i>	0.0007** (0.000)	0.0007** (0.000)	0.0031** (0.001)
<i>L.TFP</i>			0.1336*** (0.051)
_cons	1.0845*** (0.019)	1.0875*** (0.019)	0.7564*** (0.057)
AR (1)			−7.18**
AR (2)			−1.41
Sargan test			416.38
Hansen test			253.01
Year FE	✓	✓	✓
Industry FE	✓	✓	✓
Firm FE	✓	✓	✓
City FE	✓	×	✓
Year#Industry	×	✓	✓
N	14616	14616	10424
R ²	0.9703	0.9704	0.9635

development, industrial structure, and market size of different cities may have an impact on the business conditions of firms, and by controlling for city fixed effects, these region-specific effects can be excluded to more accurately assess the impact of AI on firm productivity. Column (2) in Table 4 presents the estimated results of the interaction effects controlling for industry and year. Controlling for industry and year interaction effects reduces omitted variable bias due to unobserved industry- and time-specific factors, which helps to improve model credibility and makes the findings more robust and reliable. It can be seen that the estimated results of the fixed effects model using different dimensions pass the test, and the coefficients and significance of the effect of AI on productivity do not change significantly. Further, we consider the possible endogeneity problem of the model. The endogeneity problem is a common challenge in business research, where certain variables may be correlated with the error term, leading to biased OLS estimates. GMM can solve the endogeneity problem by instrumental variables, which effectively controls the bias caused by endogeneity and improves the accuracy and consistency of estimation. Since firms' productivity is likely to be affected by the previous period, we propose to re-estimate the effect of AI on productivity by using the system generalized moments estimation model (SYS-GMM). This is because the SYS-GMM model is able to estimate the parameters of the dynamic panel data model more accurately and reduce the bias of parameter estimation compared to the traditional panel data estimation methods. By introducing instrumental variables and controlling time series correlation, the SYS-GMM model is able to improve the accuracy and robustness of parameter estimation, making the results of the study more credible. Column (3) reports the estimation results for SYS-GMM, and our findings still hold after overcoming the endogeneity problem.

5.4. Heterogeneity test results

Different forms of ownership may affect the production decisions, management mechanisms, resource allocation, and business performance of firms to different degrees. Thus, columns (1) through (3) in Table 5 report the heterogeneous effects of state-owned enterprises (SOEs), non-state-owned enterprises (non-SOEs), and Sino-foreign joint ventures, respectively. The productivity-enhancing effect of AI is greater in SOEs compared to non-SOEs and Sino-foreign joint ventures. On the one hand, SOEs are likely to be supported by government policies that encourage them to actively adopt AI technologies to improve productivity and technology [72]. In contrast, non-SOEs and Sino-foreign joint ventures may be more influenced by market competition and profitability goals, and may be more cautious about investing in technology. On the other hand, SOEs may have relatively stable and centralized organizational structures and decision-making mechanisms, and are

Table 5
Heterogeneity test.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Stated- owend	Non-Stated- owend	Joint venture	Strong international perspective	Weak international perspective	Strong sense of innovation	Weak sense of innovation
AI	0.0008* (0.000)	0.0007* (0.000)	0.0003 (0.001)	0.0014*** (0.000)	0.0002 (0.000)	0.0011* (0.001)	0.0007* (0.000)
_cons	1.1007*** (0.033)	1.1230*** (0.027)	1.0839*** (0.070)	1.1562*** (0.033)	1.0567*** (0.025)	1.1294*** (0.044)	1.0892*** (0.025)
Year FE	✓	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓
Obs.	4655	8595	1298	6022	8266	3803	10224
R ²	0.9765	0.9600	0.9779	0.9787	0.9698	0.9715	0.9727

able to make and implement decisions more quickly to promote the application and implementation of AI technologies. In contrast, non-SOEs and Sino-foreign joint ventures may have more flexible decision-making mechanisms, but they may also suffer from less efficient decision-making and greater difficulty in promoting them.

In addition, the international vision of leaders is very important to the impact of enterprise production, which can promote enterprise market expansion, technological innovation, resource allocation, talent management and risk management and other aspects of the enhancement, thus affecting the efficiency of enterprise production. We counted the number of leaders in firms with overseas experience and used the median as a dividing line to categorize the study sample into strong and weak categories. Columns (4) and (5) report results for leaders with strong and weak international horizons, respectively. As expected, firms with leaders with a strong international perspective show a stronger productivity dividend from AI. Leaders with an international perspective are able to understand global market trends and needs more deeply and identify market opportunities and challenges more clearly [73]. This enables them to utilize AI technology in a more targeted manner to meet market needs and provide products and services that are more in line with international market standards and trends, thereby increasing productivity.

Innovation awareness is also crucial in corporate production, and we include corporate innovation awareness as a third heterogeneous factor in our analysis. Specifically, we use the median number of design patent applications filed by firms as the dividing line to categorize the sample into two groups: strong and weak innovation awareness. Columns (6) and (7) report the results of the tests, which indicate that AI can be more effective in enhancing firm productivity in innovation-minded firms. The reason is that innovative companies actively explore and develop new application scenarios of AI technology in various fields, including production process optimization, automation control, data analysis and forecasting [74]. They are able to combine AI technology with existing business models and processes to create smarter and more efficient production methods.

5.5. Mechanism test results

Columns (1)-Column (3) in Table 6 report the results of regressions in which the three mechanism variables are substituted into model (3), respectively. Column (4) reports the results of the three mechanism variables and the dependent variable being included in model (4) at the same time. It can be seen that AI significantly reduces information asymmetry (ASY) and improves green innovation (GI), but does not have a significant impact on specialization division of labor (SD). The results in column (4) show that IA does have a negative effect on TFP and that SD can boost productivity, as expected. GI, however, fails to show actual TFP-enhancing capacity. This suggests that of the three mechanisms we have selected, ASY is the most dominant channel through which AI

Table 6
Mechanism test results.

	(1)	(2)	(3)	(4)
	ASY	SDL	GI	TFP
AI	-0.0103*** (0.004)	0.0010 (0.002)	0.0319*** (0.007)	0.0006* (0.000)
ASY				-0.0084*** (0.001)
SDL				0.0135*** (0.002)
GI				0.0000 (0.000)
_cons	6.0713*** (0.225)	-0.0285 (0.096)	-0.7006* (0.386)	1.1341*** (0.020)
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Obs.	14608	14616	14616	14608
R ²	0.7961	0.7666	0.6816	0.9709

enhances firm productivity. As, Alesina and Zhuravskaya [61] published in the *American Economic Review*, he demonstrates that trust is the most dominant channel through which apartheid affects the quality of government, with other mechanistic variables playing a minimal role. As far as our study is concerned, ASY is the main mechanism variable that constitutes the main mechanism for AI to enhance productivity, while the role of SD and GI is negligible.

Specifically, based on the data in Table 6, we can analyze these three factors as mediators of the relationship between AI and productivity as follows. Information asymmetry (ASY): in model (1), the coefficient between AI and information asymmetry (ASY) is -0.0103 and statistically significant. This indicates that the application of artificial intelligence can significantly reduce the level of information asymmetry. In model (4), the coefficient of information asymmetry (ASY) and total factor productivity (TFP) is -0.0084, which is also significant. This implies that the reduction of information asymmetry negatively affects productivity. Thus, this suggests that AI can enhance firm productivity by mitigating the negative effects of information asymmetry. The reason is that AI technology can quickly acquire, organize and process massive amounts of data through big data analysis and machine learning algorithms, which can help enterprises more accurately understand market demand, product trends, competitors and other information, and improve the level of information they have in the transaction process, so as to more flexibly adjust their production plans and product strategies, and to improve production efficiency [75]. In summary, hypothesis 2a is confirmed.

Specialized Division of Labor (SDL): in model (2), the coefficient of AI and Specialized Division of Labor (SDL) is 0.0010, which is a positive effect but not significant. This indicates that the positive effect of artificial intelligence on specialization division of labor is yet to be exerted.

Meanwhile, in model (4), the coefficient of specialization division of labor (SDL) and total factor productivity (TFP) is 0.0135, which is significantly positive. This suggests that an increase in the division of specialization contributes to productivity, but the mediating role of the division of specialization between AI and productivity is weaker. The possible reason is that AI has not yet been able to effectively promote the specialized division of labor within enterprises in the current application scenarios. For example, AI technology may be mainly used to automate some repetitive tasks in some enterprises, with limited effect on promoting specialized collaboration and division of labor among different departments or positions. In addition, firms may introduce AI technologies without sufficiently considering how to integrate them with the goal of facilitating a specialized division of labor, resulting in AI technologies failing to realize their full potential in this regard [76]. Therefore, the mediating effect of specialized division of labor between AI and productivity is weak due to the unclear role of AI on specialized division of labor.

Green Innovation (GI): in model (3), the coefficient of AI and Green Innovation (GI) is 0.0319, which is significantly positive, indicating that Artificial Intelligence has a positive contribution to green innovation. However, in model (4), the coefficient of green innovation (GI) and total factor productivity (TFP) is not significant. This implies that green innovation fails to effectively link AI and productivity, and the mediating role of green innovation between AI and productivity is not significant. The likely reason for this is that while AI can facilitate green innovations in companies, for example by promoting the development and application of green technologies, these green innovations may not yet be fully translatable into actual productivity gains. For example, green innovation may in some cases focus more on aspects such as environmental protection and sustainable development [77], and have less of an impact on factors that directly affect productivity, such as productivity and cost reduction in enterprises. In addition, external factors such as market mechanisms and policy environments may also affect the transformation and application of green innovations, resulting in green innovations failing to play their intermediary role effectively.

6. Further analysis and discussion

6.1. Policy moderation effect

Policy factors have a profound impact on enterprise production. Policies formulated by Governments can directly or indirectly affect the cost of production, production environment, market access, competitive environment and other aspects of enterprises. Therefore, we consider two types of policies that are directly related to firms' production: supply chain innovation policies and environmental regulation policies. We select the supply chain innovation and application pilot city and enterprise policies implemented by the Chinese government in 2018 as representative of supply chain innovation policies. We first set a proxy variable for whether or not the firm is subject to a supply chain innovation policy shock. This proxy variable takes the value of 1 if the firm is located within the pilot scope of the policy, and vice versa. Next, we cross-multiplied the AI with the proxy variables as a way to divide the treatment and control groups. Columns (1) and (2) represent the results of not being affected by supply chain innovation policies and being affected by supply chain innovation policies, respectively. By comparing the coefficients of the two, we can observe the moderating effect of the policy, which is a common approach used in current empirical studies [78,79]. In column (1) of Table 7, the coefficient of AI is 0.0004, while in column (2), the coefficient of AI is 0.0013, which suggests that firms affected by supply chain innovation policies have a more significant effect of AI on their productivity. Consistent with Wei et al. [80], they demonstrate that supply chain digital transformation can enhance firms' innovation performance, with rising supply-demand matching and efficiency improvement as important transmission paths. Supply chain innovation policies can promote the adoption of advanced logistics,

Table 7

Policy moderation effect.

	(1)	(2)	(3)	(4)
	Non-Supply Chain Policy	Supply Chain Policy	Non-low carbon policy	Low carbon policy
AI	0.0004 (0.000)	0.0013* (0.001)	0.0011*** (0.000)	0.0003 (0.001)
_cons	1.0795*** (0.023)	0.9967*** (0.062)	1.0665*** (0.024)	1.1085*** (0.033)
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Obs.	9572	3478	8692	4665
R ²	0.9727	0.9866	0.9718	0.9703

inventory management and production planning systems, and intelligent management of the supply chain through AI technology. AI can analyze a large amount of data and optimize various links in the supply chain, thereby increasing productivity and reducing costs.

Further, we denote environmental regulation by low-carbon pilot city policies, identifying firms located within the pilot as a treatment group. Specifically, we set a proxy variable for whether or not the firm is subject to a low-carbon policy shock. This proxy variable takes the value of 1 if the firm is located within the scope of the policy, and 0 otherwise. Next, we cross-multiply AI with a proxy variable for low-carbon policy shocks as a way of dividing the treatment and control groups. Columns (3) and (4) show the cases of not being affected by low carbon policies and being affected by low carbon policies, respectively. By comparing the coefficients of the two, we can observe the moderating effect of the low-carbon pilot policy. In column (3), the coefficient of AI is 0.0011, while in column (4), the coefficient of AI is 0.0003, suggesting that low-carbon policies weaken the contribution of AI to firm productivity growth. The reason for this is that low-carbon pilot city policies may require firms to adapt production processes, technologies and equipment to low-carbon requirements. For companies that have already invested a lot of money and time in the adoption of AI technology, the requirement to adapt the technology may face greater difficulties and costs, especially if the technology transformation cycle is long, which can weaken the incentive for companies to adopt AI technology. Our findings are corroborated by Ref. [81], who find that low-carbon pilot city policies are not effectively binding at the firm level, due to policy implementation bias.

6.2. AI's green development impacts

In the mechanism test, we initially demonstrated the positive impact of AI on independent green innovation. Since AI has the ability to cascade big data, curiosity drives us to think further about the impact of AI on joint-type green innovation. Therefore, we counted the number of patent applications from the Green Patent Research Database (GPRD) in three categories: green patent co-applications, green invention patent co-applications and green utility patent co-applications. We use these three types of patents to characterize "joint green innovation", "joint incremental green innovation", and "joint breakthrough green innovation", respectively. Table 8 columns (1)-column (3) report the impact of AI on each of the three types of green innovation in turn. Fortunately, AI contributes significantly to all three types of joint green innovations, showing a strong green development impact. Interestingly, AI has a greater impact on companies' incremental green innovation than breakthrough green innovation. Because incremental green innovations are more readily accepted and understood, they are easier to generalize to a wider range of firms than breakthrough innovations. The application of AI technology can make such incremental improvements more pervasive and feasible, thus better driving the entire industry toward green development. However, this inspires Chinese companies to

Table 8
Green development analysis.

	(1)	(2)	(3)
	<i>JGI</i>	<i>JIGI</i>	<i>JBGI</i>
<i>AI</i>	0.0383*** (0.008)	0.0288*** (0.007)	0.0185*** (0.006)
_cons	-1.1700*** (0.409)	-1.9567*** (0.367)	-0.5053 (0.326)
Year FE	✓	✓	✓
Industry FE	✓	✓	✓
Firm FE	✓	✓	✓
Obs.	14616	14616	14616
R ²	0.7107	0.7214	0.6747

strengthen AI cooperation in order to bring about disruptive and breakthrough technological revolutions, and not to be satisfied with incremental and slow innovation. As found by Ref. [82], they argue that digital technology-driven innovation from incremental to breakthrough is necessary to achieve the SDGs.

6.3. Historical impact analysis

Historical information infrastructures have had a profound and diverse impact on AI, of which the Ming Dynasty stagecoach is an interesting example. For the case of the Ming Dynasty stagecoach routes, we also use the interaction term for the moderating effect analysis. The proxy variable takes the value of 1 if the firm is located in the distribution area of the Ming Dynasty stagecoach routes, and 0 otherwise. We cross-multiplied AI with proxy variables as a way of dividing the treatment and control groups. Column (1) in Table 9 indicates firms located in the Ming Dynasty stagecoach route distribution area, and column (2) indicates firms not located in the Ming Dynasty stagecoach route distribution area. From the data, it can be seen that in column (1), the coefficient of AI is 0.0010, while in column (2), the coefficient of AI is -0.0001. This suggests that firms in the historical Ming Dynasty stagecoach distribution area have a more pronounced productivity boost from AI, while firms not in that area have no significant boost from AI. This is sufficient to account for the fact that the productivity effect of AI is positively moderated by the Ming stagecoach.

Conducting such an analysis by constructing interaction terms helps us to gain a deeper understanding of the moderating effect of historical factors on the relationship between AI and firm productivity. This suggests that the post station, as an important node of the ancient transportation network, assumed the function of transmitting letters, intelligence and knowledge. This centrality of information flow was of great significance to the dissemination of knowledge and information exchange at that time, providing the basis for the emergence and development of enterprises. Importantly, while the stagecoach itself was not a direct application of AI technology, it exemplified the human need for information transfer and processing, and inspired people to explore and imagine more efficient and intelligent ways of processing information. In a sense, the stagecoach provided inspiration for later

Table 9
Historical impact analysis.

	(1)	(2)
	Ming Dynasty Post	Non-Ming Dynasty Post
<i>AI</i>	0.0010*** (0.000)	-0.0001 (0.001)
_cons	1.0367*** (0.022)	1.1510*** (0.039)
Year FE	✓	✓
Industry FE	✓	✓
Firm FE	✓	✓
Obs.	9671	4049
R ²	0.9719	0.9696

developments in information technology and AI.

6.4. Differential productivity effects

Consider that the firm productivity base is also an important factor influencing the developmental effects of AI. Therefore, we further employ a panel quantile regression model to explore the regression results for productivity located at different quantiles. Columns (1)–(4) in Table 10 represent the effect of AI on productivity at the 25 %, 50 %, 75 %, and 95 % quantiles, respectively. It can be seen that the productivity-enhancing effect of AI development across firms does not differ significantly below the 75 % quartile. The productivity enhancement effect of AI is significantly larger when firms' productivity is distributed above 75 %. This suggests that firms with a high productivity base tend to be larger, with more production lines, customer bases and market shares. These firms can achieve greater economic gains through scale effects after the introduction of AI technology, resulting in larger productivity gains. In addition, this group of enterprises may be in a more competitive market environment where they need to continuously improve their productivity to remain competitive. As a result, high-productivity organizations are more inclined to adopt the latest technologies, such as AI technologies, to increase productivity and achieve market share growth.

6.5. Dynamic impact decomposition

Panel Vector Autoregression (PVAR) models take into account both time-series and cross-sectional features in panel data and are better able to capture the dynamic relationship between AI and productivity. By introducing lagged terms in the direction of the time series, the PVAR model can explore both short- and long-run dynamic effects between the variables [83]. We first determine the lag order to be first order by the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC) and Modified Akaike Information Criterion (HQIC) together. Then, Fig. 5 reports the dynamic impact of AI on productivity. The impulse response results show that the instantaneous shock of AI positively affects TFP and reaches its maximum in period 2, then gradually falls back by period 10. Our findings validate the initial effect of technological innovation. At the initial stage of AI technology introduction, firms may invest significant resources in technological innovation and transformation, leading to rapid productivity gains. The application of new technologies may improve productivity, optimize resource allocation, and enable firms to achieve higher levels of productivity in the short term. Over time, AI technology is gradually adopted and popularized by other enterprises, and the marginal effect of the technology gradually diminishes [84]. At the same time, the impact of AI on TFP may gradually diminish as market competition intensifies and the differentiation advantage between firms gradually shrinks.

7. Discussion

This study provides a comprehensive and in-depth analysis of the impact of AI on the productivity of Chinese firms and draws a series of meaningful conclusions. The academic and practical implications, limitations, and future research agenda are discussed below.

Table 10
Quantile analysis.

	(1)	(2)	(3)	(4)
	25 %	50 %	75 %	95 %
<i>AI</i>	0.0161*** (0.002)	0.0158*** (0.001)	0.0162*** (0.002)	0.0182*** (0.003)
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Obs.	15213	15213	15213	15213

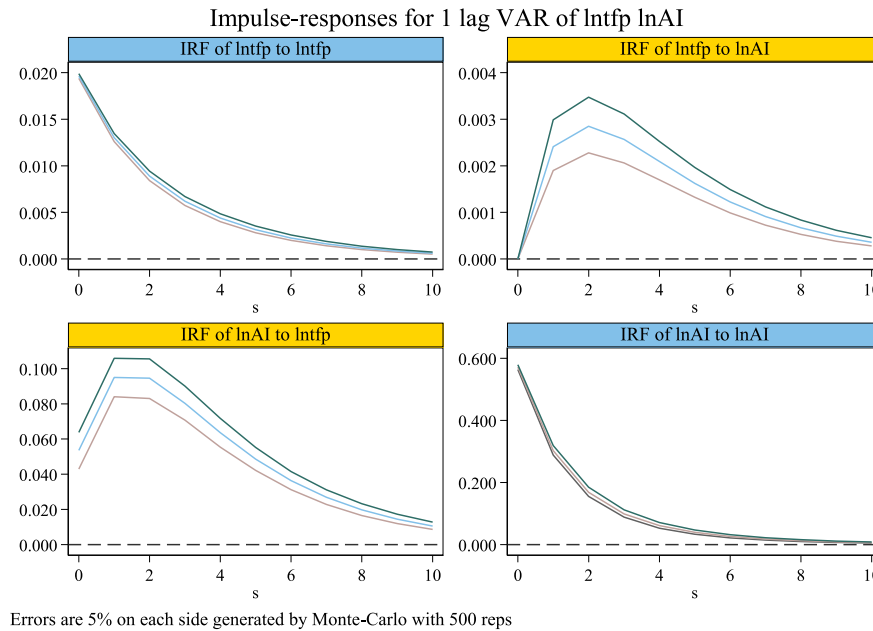


Fig. 5. Dynamic impact decomposition analysis.

7.1. Academic implications

This study sheds light on the impact of AI on firm productivity and its mechanisms at the micro level, providing new empirical evidence for research in the field of AI. Unlike previous studies that focus on the macro-level impact of ICT and digital technologies, we fill a gap in the existing literature by focusing on firm-level AI applications. For example, while Pan et al. [85] show that digital technology has a positive impact on firms' total factor productivity, our study enriches this area by further refining the specific role of AI on firms' productivity.

By verifying the direct productivity-enhancing effect of AI on firms and mechanisms such as information asymmetry mitigation, this study expands the understanding of the intrinsic mechanisms by which AI affects firm performance. This is similar to Alesina and Zhuravskaya [61]'s study, which suggests that certain mechanisms play a dominant role in a particular phenomenon, and our finding that information asymmetry is the main channel through which AI enhances firm productivity provides an important theoretical foundation for subsequent research.

The findings in this study on the moderating role of supply chain innovation policies and low-carbon pilot city policies on the productivity effect of AI, as well as the green development effect of AI, complement other studies. For example, Wei et al. [80] confirm the role of supply chain digital transformation in enhancing firms' innovation performance, and our study further reveals the moderating effect of supply chain innovation policies in the relationship between AI and firm productivity. In addition, Cheng et al. [40] found that AI enhances firms' environmental protection capabilities, and our study specifically addresses the facilitating effect of AI on firms' green innovation, especially the significant impact on joint green innovation, which provides a new perspective for research in the field of green innovation.

7.2. Practitioner implications

For enterprises, the results of this study provide clear guidance for their application of AI technology. Enterprises should recognize the important role of AI in improving productivity and realizing automated and intelligent production, and actively introduce AI technology. In particular, state-owned enterprises and enterprises with an international

perspective and a sense of innovation should give full play to their advantages and increase their investment in and application of AI. For example, Wang et al. [16] point out that the adoption of AI by enterprises can improve the efficiency of innovation, and enterprises can learn from the relevant experience and apply AI to all aspects of production to enhance their competitiveness.

Enterprises should focus on the alleviation of information asymmetry when applying AI. Through the data analysis and forecasting functions of AI, enterprises can more accurately grasp market demand and predict trends, so as to optimize production plans and marketing strategies [86]. At the same time, enterprises should also pay attention to the application of AI in quality control, personalized production and human resources optimization, giving full play to the potential of AI.

Policies formulated by governments have far-reaching impacts on enterprise production. Enterprises should pay close attention to supply chain innovation policies and actively participate in supply chain digital transformation in order to fully utilize the productivity-enhancing effects of AI. Meanwhile, for the low-carbon pilot city policy, enterprises need to respond reasonably to minimize the negative impact on AI applications while meeting low-carbon requirements. For example, Luo et al. [81] found that there are certain deviations in the implementation of low-carbon pilot city policies at the enterprise level, and that enterprises should strengthen communication with the government and jointly seek solutions in order to realize the benign interaction between low-carbon development and AI application.

7.3. Limitations

This study is mainly based on the data of listed companies in China for analysis, and the generalizability of the findings may be somewhat limited. There are differences in AI application levels and policy environments among enterprises in different countries and regions. In the future, we can consider expanding the scope of the study to include data from enterprises in more countries and regions to verify the generalizability of the findings.

In the study, AI is mainly measured by the frequency of relevant terms in annual reports, which may not be able to accurately reflect the development level of enterprise AI. The development of AI is a dynamic process, which involves the continuous updating of technology and the

expansion of application scenarios, and in the future, we can explore more accurate measurement methods, such as adopting the indicators to directly assess the level of application of enterprise AI technology.

The factors affecting enterprise productivity are numerous and complex, and although this study has considered some major factors, there may still be other factors that have not been included, such as the management level of the enterprise and enterprise culture. These factors may have some impact on the results of the study, and future research can further expand the selection of variables to reveal the relationship between AI and enterprise productivity more comprehensively.

7.4. Future research agenda

Examine the impact of the convergence of AI with other emerging technologies (e.g., IoT, big data, etc.) on enterprise productivity. The convergence of these emerging technologies brings new opportunities and challenges for businesses, and future research could focus on the synergies and interactions between them, as well as trends in different policy environments.

Focus on the ethical and social issues brought about by AI [26]. The development of AI may lead to the restructuring of employment, the protection of data privacy and other issues, and future research needs to explore these issues in depth and propose corresponding solution strategies in order to realize the sustainable development of AI.

Future research could focus on the absorptive capacity of firms adopting AI. Specifically, exploring the dynamic relationship between firms' absorptive capacity and the development of AI technology is an important research agenda. As AI technology continues to evolve, the absorptive capacity of firms will need to improve, and future research could focus on how firms can adjust and optimize their absorptive capacity in time to adapt to changes in the technology.

8. Conclusions and policy implications

AI technology has become an important engine driving social development and change, and exploring its micro-enabling role in enterprise production can help countries seize the opportunities of the AI era and realize sustainable development. This study comprehensively analyzes the heterogeneous effects and internal mechanisms of AI on the productivity of Chinese firms through a variety of research methods, and draws a number of important conclusions.

First, the innovative finding of this study that AI is effective in increasing firm productivity and that this finding remains robust after changing the research methodology provides strong evidence of the positive role of AI in firm production. Second, the study innovatively points out that the nature of enterprise ownership, international vision and innovation consciousness of leaders moderates the impact of AI on productivity, which is manifested in the fact that AI has a greater impact on the productivity of state-owned enterprises, and enterprises whose leaders have strong international vision and innovation consciousness, which provides new perspectives on enterprise management and decision-making. Again, this study finds that information asymmetry is the main mechanism by which AI enhances firm productivity, and that specialization and green innovation also show potential channel effects, a finding that expands the understanding of the mechanisms by which AI works. In addition, the fact that supply chain innovation policies positively moderated the impact of AI on productivity, while low-carbon pilot city policies had the opposite effect, provides a valuable reference for policymakers. At the same time, AI has a significant green development effect, which can significantly promote the joint green innovation capability of enterprises, especially the promotion of progressive green innovation is more obvious, which reflects the important role of AI in sustainable development. In addition, the finding that historical infrastructure development has a significant modernization effect, and that AI development in areas where historical Ming Dynasty stagecoaches were distributed increased firm productivity more

significantly, reveals an interesting link between history and modern technological development. Finally, there are differences in the impact of AI on firms with different productivity bases, with a greater positive impact on firms with productivity at the 75th percentile and above, which provides a basis for individualized growth strategies. In addition, the positive impact of AI on productivity shows a "first rise and then fall" boosting effect, which provides new insights into the pattern of AI's impact.

In summary, these innovative findings of this study provide a comprehensive and unique perspective for a deeper understanding of the role of AI in enterprise production, which will help countries and enterprises better grasp the opportunities of the AI era and realize sustainable development. Based on the above findings, we can summarize the following four policy implications:

First, the government can formulate policies to promote enterprises to accelerate the pace of digital transformation and smart manufacturing, and encourage them to apply AI technology to improve productivity and product quality. Second, the government can incentivize private and Sino-foreign joint ventures to increase investment and application of AI technology through policy instruments such as tax incentives, financial support, and innovation incentives. Finally, the government can encourage enterprises and leaders to develop a sense of innovation and an international outlook through education and training and support for scientific research projects. This includes encouraging enterprises to cooperate with leading international AI firms to absorb advanced technology and management experience, thereby improving their competitiveness and innovation.

Consider the role of information asymmetry as a mechanism in AI and productivity enhancement. The government can establish a trust mechanism and certification system for AI technology, certify and supervise AI products and services that meet the standards, improve enterprises' confidence in AI technology, and promote its application and productivity enhancement. Supply chain innovation policy has a positive regulating effect, and the government and enterprises should strengthen cooperation to jointly promote supply chain innovation, realize supply chain digital transformation and intelligent upgrading, and improve the efficiency and competitiveness of the whole industrial chain. Low-carbon pilot policies may lead to limited application of AI technology, increased cost pressure and difficulties in technology integration. Governments and enterprises can increase support and investment in AI technology, formulate more flexible low-carbon policies, and promote technology integration and innovation to realize the positive interaction between low-carbon development and AI technology application.

AI technology can promote cross-border cooperation and knowledge sharing among enterprises, especially in the field of green technology innovation. Enterprises can share data, algorithms and models to jointly develop AI applications that solve environmental problems and promote joint green technology innovation. At the same time, the government can promote the formation of an industrial ecosystem for green technological innovation and foster cooperation and innovation among enterprises upstream and downstream of the industrial chain. The construction of post stations in the Ming Dynasty helped the government to collect and transmit information from all over the world, providing support for decision-making. Similarly, in modern society, data collection and sharing are crucial for the development of AI technology. The government and enterprises can establish a data-sharing platform to facilitate the sharing of data resources among all parties to support the application and development of AI technology.

Firms with high productivity levels are likely to focus more on optimal allocation of resources and efficiency gains, and are therefore more willing to invest in AI technologies to smarten and automate their production processes. When formulating relevant policies, the government can focus on the needs and characteristics of this part of the enterprises, and formulate targeted support policies to provide them with more support and encouragement, so as to promote the wide application

and in-depth development of AI technology in these enterprises. In addition, considering that the impact of AI on productivity will gradually slowdown in the long run. Initially, the government can encourage enterprises to increase their investment in AI technology by providing financial support and tax incentives. With the application of the technology, the government can adjust its policy focus in due course, focusing on technological innovation and management optimization to promote industrial development to a higher level.

CRedit authorship contribution statement

Zhiyao Sun: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation. **Shuai Che:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jie Wang:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Data curation, Conceptualization.

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Data availability

Data will be made available on request.

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